Tribhuvan University
Institute of Science and Technology
2071

Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology Stat. 103
(Probability and Statistics)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
The figures in the margin indicate full marks.
All notations have the usual meanings.

Group A

Attempt any Two: (2×10=20)

1. Show that the mean and variance of Poisson distribution and equal. Telephone calls enter a college switchboard on the average of two for every 3 minutes. If one assumes an approximate Poisson process, what is the probability of three or more calls arriving in a 9-minute period?

2. In a certain factory, three machines A_1, A_2 and A_3 make 30%, 45% and 25% respectively, of the products. It is known from past experience that 2%, 3% and 2% of the products made by each machine, respectively, are defective. If a product were chosen randomly and found to be defective, what is the probability that it was made by machine A_3?

3. A chemical company, wishing to study the effect of extraction time on the efficiency of an extraction operation, obtained the data shown in the following table:

Extraction Time in minute (X)	27	45	41	19	35	39	19
Extraction efficiency in % (Y)	57	64	80	46	62	72	52

   (a) Fit a straight line to the given data by the method of least squares and use it to predict the extraction efficiency one can expect when the extraction time is 35 minutes.

   (b) Determine the coefficient of determination and interpret its meaning.

Group B

Answer any eight questions: (8×5=40)

4. The probability that an integrated circuit chip will have defective etching is 0.12, the probability that it will have a crack defect is 0.29, and the probability that it has both defects is 0.07. What is the probability that a newly manufactured chip will have either an etching or a crack defects.

5. Differentiate between point estimation and interval estimation. Explain the properties of a good estimator.

6. The following data represent the total fat for burgers from a sample of fast-food chains.
   19, 31, 34, 35, 39, 40, 35, 43, 46, 50
   Compute mean, median and mode, and on the basis of these descriptive measures, explain the nature of the data.

7. The joint probability density function of two random variables (X, Y) is given by

$$f(x, y) = 2, \quad \text{for } 0 < x < 1, 0 < y < x$$
$$0 \quad \text{elsewhere}$$

(i) Find the marginal probability density function of X and Y

(ii) Find the conditional probability density function of Y given X given X = x, and conditional probability density function of X given Y = y

8. Given a random variable having the normal distribution with $\mu = 30$ and $\sigma^2 = 25$. Find the probabilities that (i) $26 \leq X \leq 40$ (ii) $X \geq 45$ (iii) $|X - 30| > 5$.

9. Define Chi square distribution and its density function. For what purposes Chi-square test is used? Explain.

10. Obtain the maximum likelihood estimate for the parameter P (proportion of success) of binomial distribution.

11. The average zinc concentration recovered from a sample of zinc measurements in 36 different locations is found to be 2.6 grams per milliliter. Find the 95% confidence interval for the mean zinc concentration. Assume that the population standard deviation is 0.3.

12. Define regression analysis. State the simple linear regression model, and explain its assumptions.

13. Write short notes on the following:

a) Characteristics features of binomial distribution

b) Properties of Karl Peason's correlation coefficient